Research Statement
ICL-Guided Navigation of the Neural Thicket

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1 Motivation and Problem

There is a gap between what a large language model knows and what it can do on any given task. A pre-trained model is optimised to be useful on average across an enormous distribution, which means it is rarely optimal for anything specific. Closing that gap currently requires fine-tuning, careful prompting, or retrieval, all of which treat the base model as a fixed artifact to work around rather than a structure to understand.

A recent result by (Gan and Isola, 2026) reframes this problem in a way I find compelling. They show that within a small neighbourhood of a pre-trained model in weight space, there already exist highly specialised networks, these adjacent models that substantially outperform the base on individual tasks without having been trained for them. The catch is that finding these *task specialists* currently relies on random Gaussian perturbation: sample noise, evaluate, repeat. This works in principle, but scales poorly. The neighbourhood is high-dimensional, and isotropic noise has no reason to prefer the directions where task-relevant capabilities actually vary.

The question I want to pursue is whether we can do better using what the model itself tells us. Work on LoRA and related methods (Hu et al., 2022) has shown that task-specific fine-tuning concentrates in low-rank subspaces of the weight matrix. Telling us that the updates are not random, they are structured. Separately, in-context learning (ICL) research (Brown et al., 2020; Dherin et al., 2025) has shown that a handful of demonstrations can activate task-relevant circuits without changing any weights at all. Together, these suggest that demonstrations carry implicit directional information about where in weight space a specialist might live. If that information can be extracted and used to bias the search, the Thicket becomes navigable rather than just explorable.

2 Proposed Research

The central proposal is to replace isotropic sampling with ICL-guided MCMC. Given a base model θ_0 and a target task described by demonstrations D, a forward pass with D produces a gradient signal w.r.t. θ_0 that identifies which attention heads and MLP circuits are most activated. This gradient defines a low-rank subspace that can serve as the mean direction for a Metropolis–Hastings proposal over weight perturbations δ, with task performance on a held-out set acting as the un-normalised log-likelihood. The sampler walks toward specialists rather than diffusing randomly.

The rotation project would test whether this actually helps. The comparison is straightforward: ICL-guided MCMC versus isotropic sampling, evaluated on tasks from reasoning (Countdown, GSM8K, MATH-500, OlympiadBench), code generation (MBPP), and chemistry (USPTO), run on open-weight models in the 7B–13B range. The metrics that matter are how many evaluations it takes to reach a target accuracy, how good the resulting specialist is, and whether it holds up under distribution shift within the task class.

If the guided sampler works, several follow-on questions become interesting. One is whether specialists found for related tasks sit on a low-dimensional manifold; if so, interpolating between them might give cheap compositional generalisation. Another is whether the walk can be resumed for new tasks without losing what it has already found, which connects directly to the continual learning problem (Kirkpatrick et al., 2017). A third is mapping the boundary of what is reachable from a given base model, which has practical implications for capability auditing. All of these are downstream of the core question, so the rotation result would shape which direction is worth pursuing.

3 Background

I came to this problem from two directions that do not often meet. My background in probabilistic methods (MCMC, variational inference, Bayesian modelling) gave me the machinery, and a growing interest in how large models represent and process information gave me the motivation to use it somewhere that seemed underexplored.

The clearest expression of that combination so far is my first-author paper at NeSy 2026, which came out of a three-month research visit to the University of Edinburgh under Prof. Vaishak Belle (September–November 2025). That work introduced a Prototype-Guided MCMC over symbolic solution spaces, using structured prototypes to steer the sampler away from reasoning shortcuts. The connection to what I am proposing here is not superficial. In both cases, the idea is to replace an uninformed sampler with one that has a structured prior derived from the problem itself. Working through that paper convinced me the approach is technically viable and that the weight-space version of the same idea is a natural next step.

Bibliography

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